

Choose Informative Questions

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Verify the candidate scores for Fitting a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

6. Solve the given case

Eight blue cards contain four circles and four triangles; target is shape.

Which question is more useful: blue or circle?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 15 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Ask first about preferences that remove the largest number of unsuitable books.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Name the parts and decisions

Identify the root, its two branches, leaf predictions and depth of the worked fitted tree. Which parts were selected using training labels?

My reasoning and evidence

13. W2 Verify the candidate scores

For each near branch and each tall branch, count blocked and clear labels and choose a leaf. Give the total errors for each candidate question.

My reasoning and evidence

14. W3 Separate fitting from prediction

A learner sees a blocked validation case with $\text{near}=0$ and changes that leaf to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

15. W4 Fit a new tree independently

New training rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree predictions and compare accuracy with the training-majority baseline.

My reasoning and evidence

16. W5 Audit a tie and a claim

If near and tall make the same training error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Choose Informative Questions

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

Worst-case remaining candidates

cards remaining



Bar length shows the supplied value; the marked baseline is zero.

Choose Informative Questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. List possibilities

There are eight possible target cards initially.



2. Mystery question

Yes group 4; No group 4; largest remaining group 4.



3. Paperback question

Yes 6; No 2; largest remaining group 6.



4. English question

Yes 8; No 0; largest remaining group 8.

CHANGE ONE THING

The current possibility set contains only the four mystery cards.

Explain the changed result using the matching lesson.

Choose the best original question and explain why its more balanced split matters here.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Yes 8; No 0; largest remaining group 8.

1. List possibilities
2. Mystery question
3. Paperback question
4. English question

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: English question Yes 8; No 0; largest remaining group 8. Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set. This is an identification-game criterion; a classifier fitting labels may instead compare prediction errors or another declared training measure.

Practice 2: explain the deciding evidence

Changed-case practice: The current possibility set contains only the four mystery cards. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Asking mystery now leaves all four; it no longer narrows the set.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Asking mystery now leaves all four; it no longer narrows the set. Question value depends on the remaining candidates, not just on its wording.

Repair a missing building block

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf predictions from training examples.

Short practice: Why does changing colour alone not change this tree's prediction?. Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?. Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
